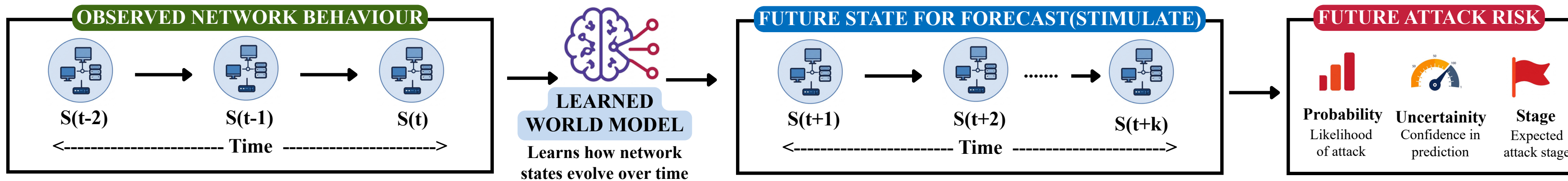


- **Problem Statement ID** : SIH26153
- **Problem Statement Title** : AI based Network Attack  
Forecasting from Network Traffic Data
- **Theme** : Blockchain & Cybersecurity
- **PS Category** : Software
- **Team ID** :
- **Team Name (Registered on portal)** :



## PROPOSED SOLUTION : From detection to *Future-state forecasting*



### UNIQUE VALUE PROPOSITION (USP)

- **Leakage-tested at every layer**  
timing bias, episode memorization, and split leakage are independently verified absent, not assumed.
- **One model, two signals**  
forecasts future attacks and flags unseen behaviour.
- **Counterfactual defence simulation**  
tests how the future attack trajectory changes when the defender acts.

### Detection-Focused systems



Identify Suspicious activity from current or recent observations



### Our Approach : Forecasting - focused Systems



Learn how network states evolve and stimulate possible future states

### HOW IT ADDRESSES OUR PROBLEM

- **Early Forecasting:** Predicts attack risk **before** the attack occurs.
- **Future-State Prediction:** Forecasts how **network behaviour** will evolve.
- **Attack Progression:** Predicts the **likely next attack stage (K-steps)**.
- **Uncertainty-Aware:** Reports **probability** and **prediction confidence**.
- **Behaviour-Based:** Uses **network traffic behaviour**, not only fixed attack signatures.



## TECH STACK

### Core

Python 3.10 - pandas / PyArrow / NumPy -  
PyYAML - Scapy

### Dataset

CSE-CIC-IDS2018 (primary) - CTU-13  
(exploratory)

### Modeling

PyTorch (LSTM + dynamics head) -  
PyTorch Geometric (GraphSAGE) - scikit-learn  
(baselines)

### Evaluation

Custom LOEO + chronological split - scikit-  
learn metrics - Matplotlib - pytest

### Demo

Streamlit (offline, single process) - Plotly  
charts - predict() inference contract — no  
REST API

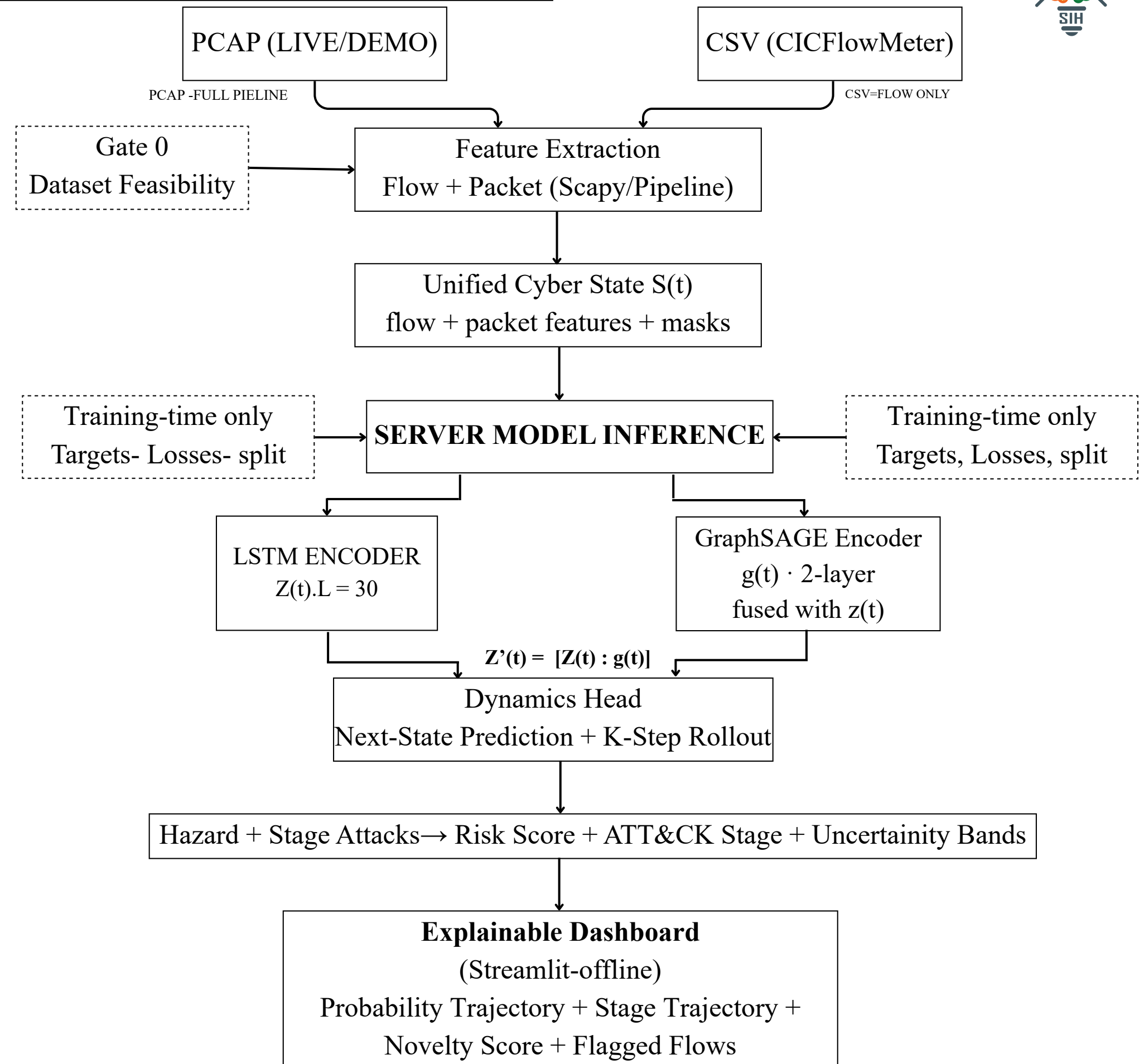
### Explainability

Feature-group ablation + deletion validation

### Infrastructure

Fully offline - local filesystem only - Git +  
frozen configs + SHA256 - no cloud / no  
containers

## TECHNICAL APPROACH



## FEASIBILITY:



### OPEN DATA & OPEN-SOURCE

Public datasets + open-source tools

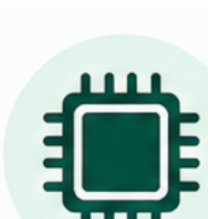
*No proprietary software dependency*



### OFFLINE DEPLOYMENT

No cloud or internet dependency

*Suitable for air-gapped environments*



### EFFICIENT INFERENCE

Offline training + lightweight runtime

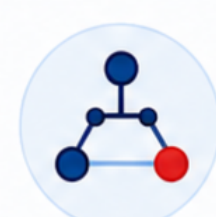
*Designed for timely forecasting*

## VIABILITY



### DATASET SCHEDULING ARTIFACTS

Short-horizon forecasts may be influenced by attack timing.



### RARE MULTI-STAGE ATTACK EXAMPLES

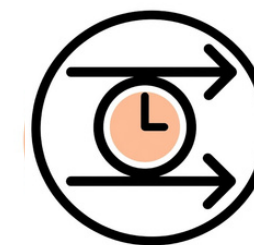
Limited labelled progression / lateral-movement episodes.



### AUTOREGRESSIVE ERROR GROWTH

Prediction uncertainty compounds over longer rollouts.

## OUR STRATEGY



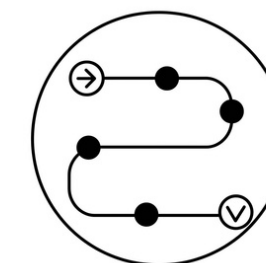
### SCHEDULE CONTROL TESTING

Separates genuine forecasting from timing effects.



### CASE-STUDY VALIDATION

Reports stage predictions only where evidence is sufficient.



### VALIDATED FORECAST HORIZONS

Measures reliability and reports only supported horizons.

| WHO                                 |                                                                                     | IMPACTS                                        | BENEFITS                            |
|-------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Organizations                       |    | Forecast attacks before major compromise       | Reduced financial losses            |
|                                     |                                                                                     | Estimate potential operational impact          | Improved business continuity        |
| Security Teams                      |    | Prioritize threats using predicted risk        | Faster threat investigation         |
|                                     |                                                                                     | Identify the features driving each prediction  | Reduced alert overload              |
| Network Administrators              |  | Track changes in network behaviour over time   | Better network visibility           |
|                                     |                                                                                     | Observe predicted future network states        | Proactive infrastructure management |
| Critical Information Infrastructure |  | Anticipate threats to essential services       | Improved service resilience         |
|                                     |                                                                                     | Assess possible attack progression             | Reduced risk of service disruption  |
| Security Decision Makers            |  | Compare risk across future attack trajectories | Better resource allocation          |
|                                     |                                                                                     | Support risk-based security decisions          | More informed security planning     |

## REFERENCES:

- CSE-CIC-IDS2018 Dataset, Canadian Institute for Cybersecurity (CIC), University of New Brunswick.
- MITRE ATT&CK Framework, MITRE Corporation.
- CAPEC (Common Attack Pattern Enumeration and Classification), MITRE Corporation.
- National Vulnerability Database (NVD), National Institute of Standards and Technology (NIST).
- Ha, D. & Schmidhuber, J. (2018). World Models.
- Engelen, G. et al. (2021) – Research on CICIDS dataset labeling and preprocessing.
- Lanvin, M. et al. (2023) – Improvements to CICIDS flow construction and labeling.

## LINKS:

- <https://www.unb.ca/cic/datasets/>
- <https://attack.mitre.org>
- <https://capec.mitre.org>
- <https://nvd.nist.gov>